



Improved data interoperability for GRNsight: a web application for visualizing models of gene regulatory networks

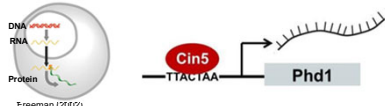
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<http://dondi.github.io/GRNsight/>

Gene Regulatory Networks (GRNs) Can Be Illustrated by Directed Graphs

- The central dogma of molecular biology describes how the flow of information in a cell during gene expression goes from DNA to RNA to protein.
- Transcription factors control gene expression by binding to regulatory DNA sequences.
- Activators increase gene expression.
- Repressors decrease gene expression.
- Transcription factors are themselves proteins encoded by genes.



- A gene regulatory network (GRN) consists of genes, transcription factors, and the regulatory connections between them, which govern the level of expression of mRNA and proteins from those genes.
- Each node in a GRN graph represents the gene, mRNA, and protein expressed from the gene; each edge represents a regulatory relationship.
- All the nodes are transcription factors themselves.

GRNsight Fulfills a Specific Software Niche for Visualizing Small- to Medium-scale GRNs

- Although other open source software, such as Cytoscape or Gephi, exists to lay out large networks, they were too cumbersome for our needs.
- GRNsight is targeted at both experienced biology investigators and novice undergraduate users and has the following requirements:
 1. Exist as a web application.
 2. Be simple and intuitive to use.
 3. Accept Excel (.xlsx) files directly from our sister project, GRNmap, as well as, SIF (.sif), or GraphML (.graphml) input files.
 4. Read a weighted or unweighted adjacency matrix.
 5. Automatically lay out and display unweighted and weighted, directed network graphs.

GRNsight Accepts Microsoft Excel, SIF, and GraphML Files in the Proper Format

Excel

- Excel workbooks need a "network" sheet (for unweighted graphs) or a "network_optimized_weights" sheet (for weighted graphs). The adjacency matrix can be symmetrical or asymmetrical.
- GRNmap input and output workbooks are accepted without adjustment.
- Adjacency matrices generated from other databases, such as YEASTRACT (Teixeira et al., 2014), can be used with some modification.

GRNsight now accepts SIF and GraphML files without any modification

SIF

- A SIF file is a tab-delimited text file with the file extension .sif originally created for use with Cytoscape.
- Lines in the SIF file specify a source node, a relationship type (or edge type), and one or more target nodes separated by tab characters.

GraphML

- A GraphML file is an Extensible Markup Language (XML) file with the extension .graphml.
- GRNsight parses GraphML to extract nodes and edges.
- GraphML has the ability to specify additional graph features that GRNsight cannot display, but may be useful additional data.

GRNsight Automatically Lays Out Unweighted and Weighted Network Graphs

Excel Spreadsheet

1. Menu Bar

- Four demo files can be found in the "Demo" menu for users who do not have their own data.
- Print functionality is accessed from the "File" menu option.
- "File > Reload" reloads the current graph with the active settings.
- In the "Edit > Preferences" menu, the user can select for weighted graphs to be displayed as unweighted graphs.

2. Force Graph Parameter Sliders

- Link distance determines the minimum distance between nodes.
- Nodes have a charge, which repels or attracts other nodes.
- The charge distance determines at what range a node's charge will affect other nodes.
- Gravity determines the strength of the force holding the nodes to the center of the graph.
- Reset functionality sets all parameters to default settings.
- Locking the parameters prevents any further changes.

3. Nodes, Edges, and Arrows

- The absolute values of each weight parameter are normalized to a value between 0 and 1. The thicknesses of the lines are adjusted to vary continuously from the minimum thickness (for normalized weights near zero) to the maximum thickness (normalized weights of 1).
- Edges with negative weights (repression) are colored cyan with blunt end markers; edges with positive weights (activation) are colored magenta with pointed arrowheads; edges with normalized weight values between -0.05 and 0.05 are colored grey to signify a weak influence on the target gene.

4. Status Bar

- The name of the uploaded file is displayed.
- The number of nodes and edges of the graph are displayed in the far right hand side.

3. Weighted graph laid out automatically

Produced in ~10 milliseconds

Unweighted graph drawn manually with Adobe Illustrator

Produced in several hours

Unweighted graph laid out automatically by GRNsight and adjusted by hand

Produced in ~5 minutes

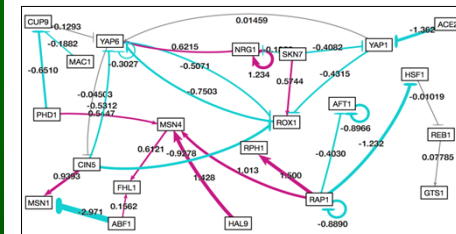
Weighted graph laid out automatically by GRNsight and adjusted by hand

Produced in ~5 minutes

New Visualization Features

GRNsight Beta includes options to show or hide the weight values

- Radio buttons were added to the main toolbar to the left of the graph bounding box that enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover of the edges.
- Achieved by leveraging CSS.



GRNsight Beta also includes bounding box resizing

- Radio buttons allow the user to change the size of the bounding box, and thus the size of the graph.
- Users can also zoom in and out of the visualization, and, if enabled, can scroll around within the bounding box.
- This set of features was coded via D3.js, the main visualization component in GRNsight.

Future Directions

- Further expand and refine the unit testing framework.
- Allow users to set edge thickness normalization values so that different graphs can be compared on the same scale.
- Calculate and present graph statistics, such as betweenness centrality.
- Add new graph layout options.

Availability

- GRNsight is free and open to all users and there is no login requirement.
- Web site content is available under the Creative Commons Attribution Non-Commercial Share Alike license.
- GRNsight code is available under the open source BSD license.
- Usage is being tracked through Google Analytics.
- GRNsight has been tested with and confirmed to be working in Chrome version 43 or higher and Firefox version 38 or higher on Windows 7 and Mac OS X.



Acknowledgments

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